LED downlights in Insulspan roof panels

How to fit LED downlights into a Quick Built insulated roof panel: the light, where it can go, the install steps and how it is powered.



LED downlights in a Quick Built Insulspan patio ceiling.

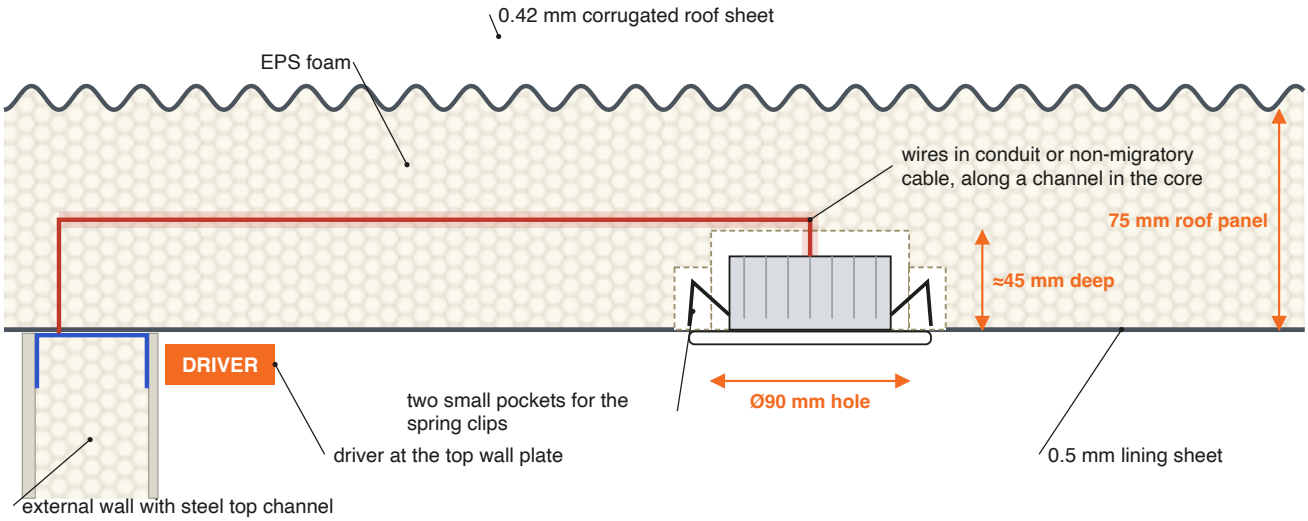
The light

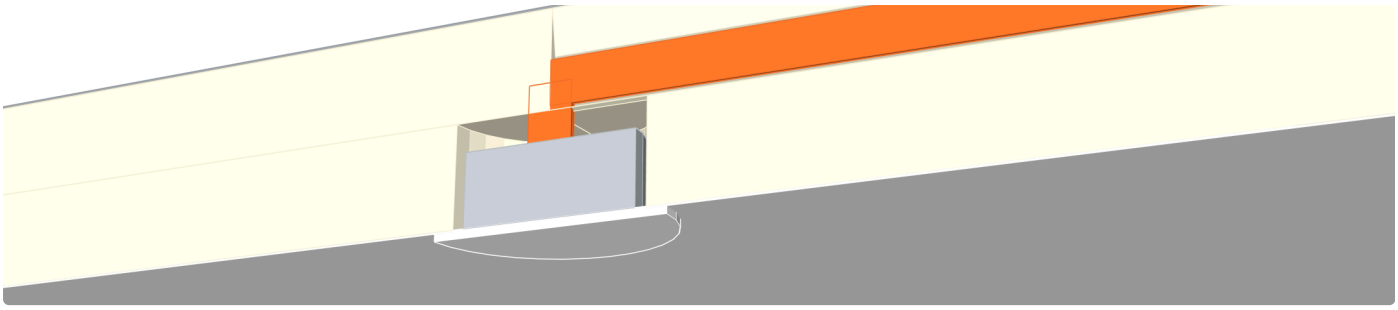


Cut-out	Ø90 mm
Light depth	Slimline, about 25–33 mm
Rating	IC-4 and AC-F, IP44, dimmable
Heat	Low-heat light made for insulated roof panels
Driver	Separate driver: how it is placed depends on the lead length (section 4)

All electrical work must be done by a licensed electrician to AS/NZS 3000. Plan the lights and their cables before the roof goes on: the rough-in is done as each panel is installed.

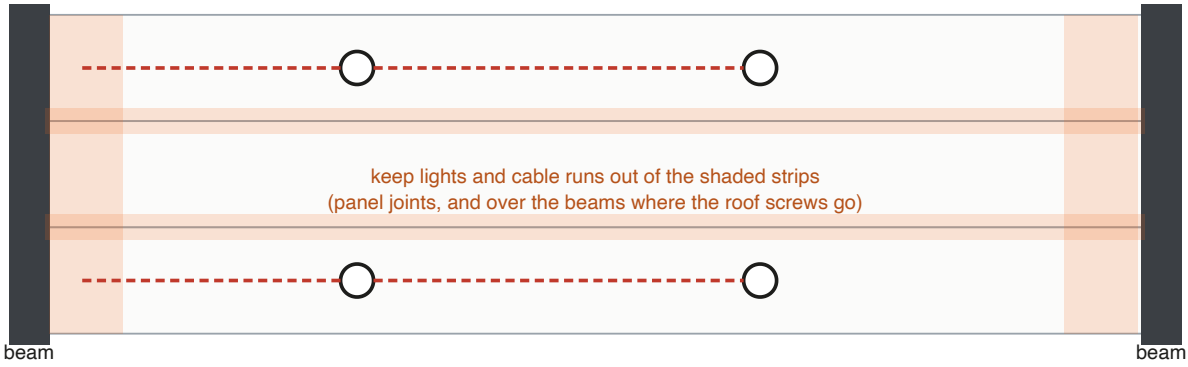
1 How it sits in the panel





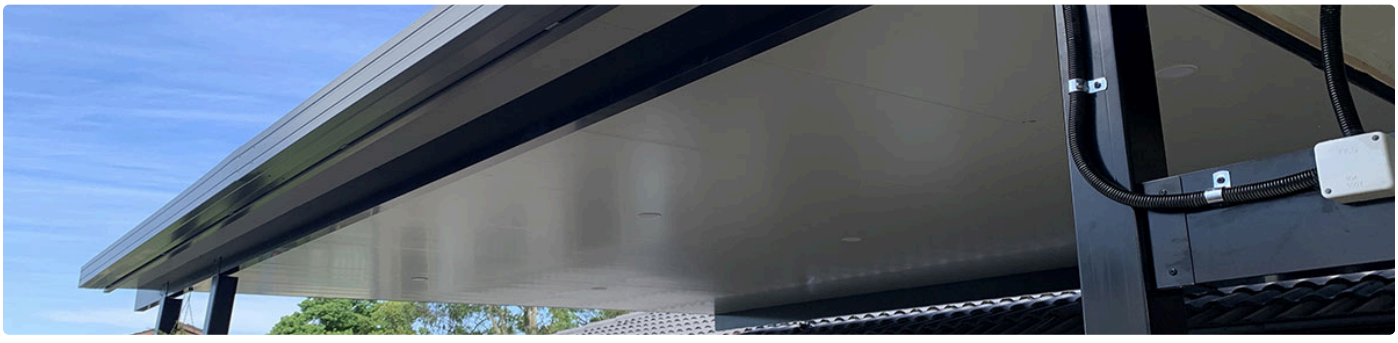
3D cutaway: the light in its Ø90 hole, with the cable in conduit along a channel in the foam.

2 Where the lights can go



3 Install steps

1. **Mark each light** on the ceiling, away from the panel joint, and with all services clear of the top plate / beam where the roof screws go through. Leave the ceiling's protective film on while marking, drilling and cutting.
2. **Run the cable** to each light as the panels go up, along a service channel in the panel: at the panel edge joint where the panel has one, or a channel made with a long auger bit or by carefully pushing a hot reo bar through. Then drill sideways through the foam to the light hole.
3. **In the foam**, the wires must run in conduit or be a non-migratory cable.
4. **Cut the hole** with a Ø90 mm metal-cutting hole saw through the ceiling, to a depth of about 45 mm. Take extra care in 50–65 mm panels.
5. **Carve out two small sections** of foam either side of the hole so the spring clips sit flush against the inside face of the steel ceiling.
6. **Place the driver** as in section 4: outside the panel on a long low-voltage lead, or a smaller driver beside the light. Electrical parts inside the panel must stay under 80 °C.
7. **Seal any wrongly drilled holes** and clean off the swarf.
8. **Peel the protective film** off the ceiling.

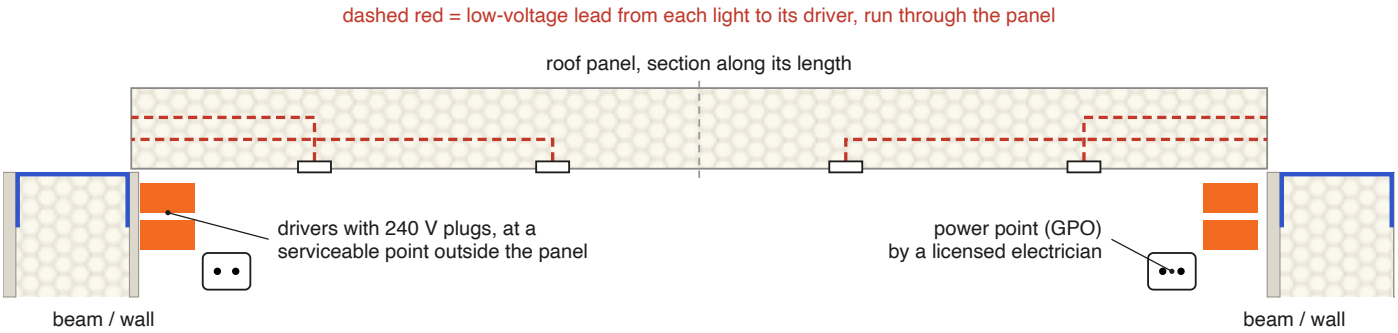


Cable brought down from the roof in flexible conduit to a junction box, on a Quick Built Insulspan patio.

4 Power to the lights: plug-in drivers and long panels

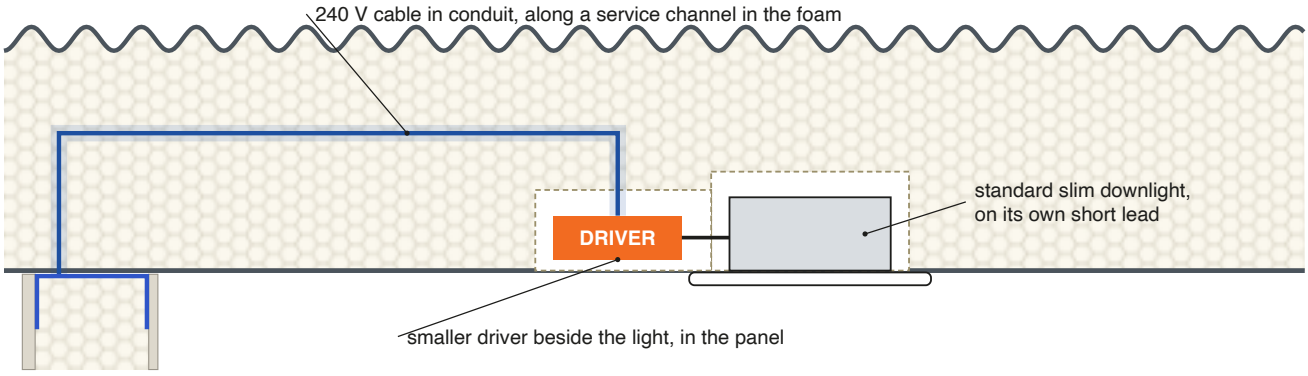
A downlight has two parts: the light, and a driver that plugs into 240 V. How far apart they can be depends on the lead between them. **Do not cut the light's lead and join it inside the panel:** every cable joint must stay accessible, inside an enclosure.

Method A: light with a long low-voltage lead (driver outside the panel)



- Use a downlight sold for insulated roof panels: these come with 10–12 m of low-voltage lead, or with plug-together low-voltage extension leads (6 m and 10 m lengths are sold).
- The low-voltage lead runs through the panel. The driver and its 240 V plug sit outside the panel at a serviceable point: in the roof cavity of an attached house, behind the receiver channel on a freestanding patio, or on the top wall plate.
- The power point (GPO) for the plugs is installed by a licensed electrician to AS/NZS 3000, and can be put in before the roof panels go on.

Method B: standard slim downlight with a short lead (driver inside the panel)



- The electrician runs the 240 V cable in conduit along a service channel in the panel, then through the foam to the light hole, as each panel is installed.
- A smaller driver is fitted inside the panel beside the light, with the light's own short lead. No joins are buried in the foam: the connection is at the light hole.
- Electrical parts inside the panel must stay under 80 °C.

A common spacing is one light every third panel.